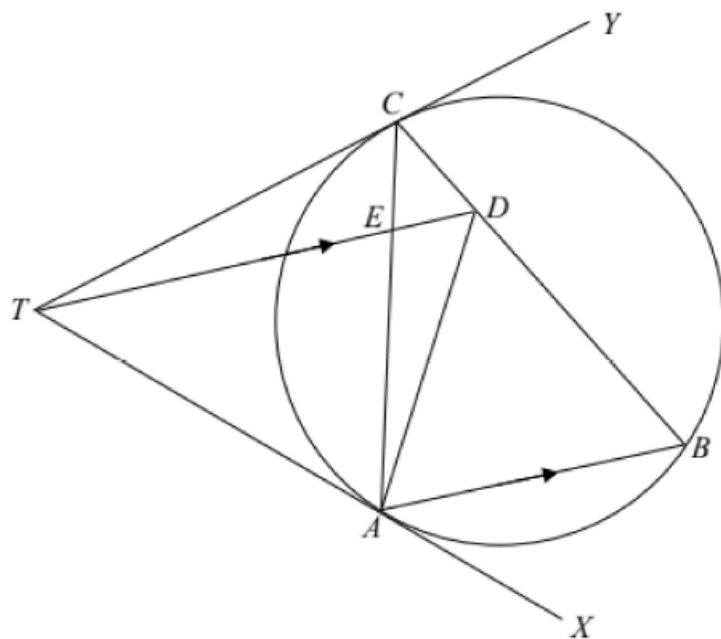


1	ACSB/2025/I/12
	In the diagram, ABT is a straight line. CT is the tangent to the circle at C. Prove, giving all reasons, that $CT^2 = AT \times BT$. [5]
2	BASS/2025/I/07
	The diagram shows points A, B, C and D lying on the circle. The point E is such that DCE is a straight line and EBF is a tangent to the circle at point B. AB is parallel to DE and $AB = DB$. (a) Prove that angle $CBE =$ angle ABD. [2] (b) Prove that triangle CBE is similar to triangle ABD. [3]

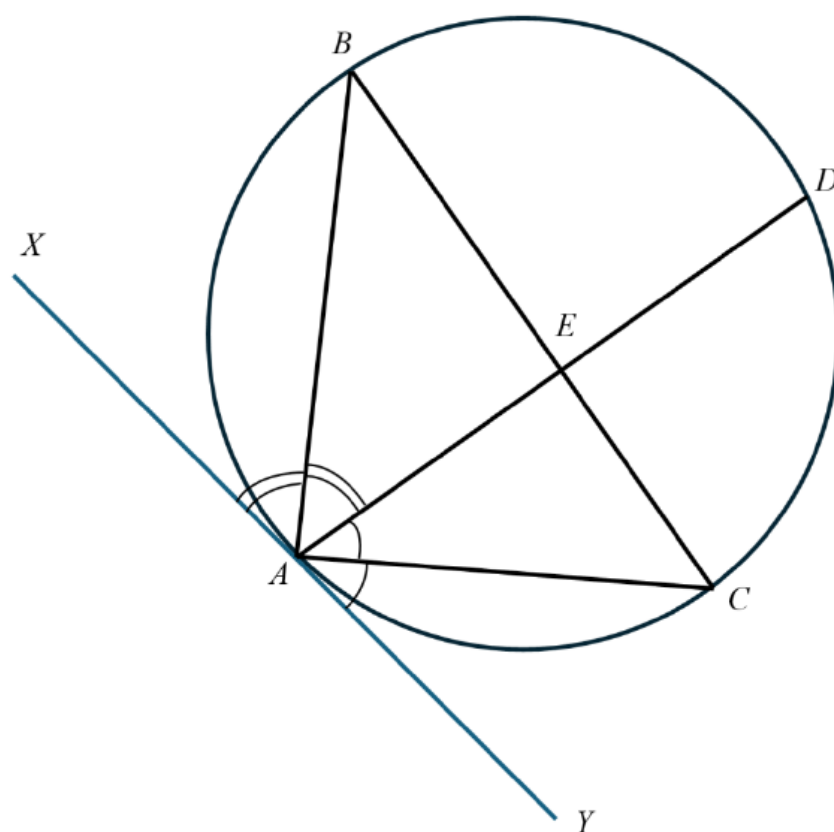
3

CCHY/2025/I/6

In the diagram below, TAX and TCY are tangents to the circle at A and C respectively. AC meets TD at E . D is on BC such that TD is parallel to AB .



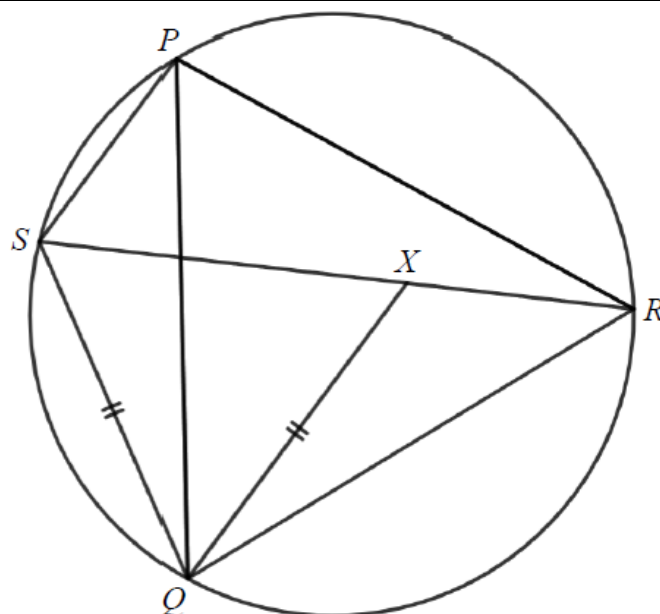
- (i) Prove that angle ACB is equal to angle ATD . [2]
- (ii) Explain why a circle can be drawn passing through the points T, A, D and C . [1]
- (iii) Hence prove that $CE \times AE = DE \times TE$. [3]



The diagram above shows a circle passing through the points A , B , C and D . The line XAY is a tangent to the circle at point A .

AED is a straight line such that AC bisects angle DAY and AB bisects angle DAX .

- (a) Show that triangle BAC is similar to triangle AEC . [3]
- (b) Show that $AC^2 = EC \times BC$. [1]
- (c) Show that BC is a diameter of the circle. [3]

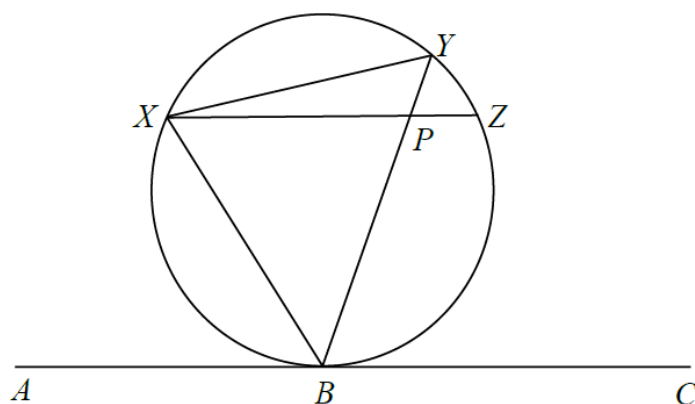


In the diagram, PQR is an equilateral triangle inscribed in a circle.
 S is a point on the circumference of the circle.
 X is a point on the line SR and $QS = QX$.

Prove that triangle RXQ is congruent to triangle PSQ .

[6]

In the diagram, ABC is a tangent to the circle at point B . The points X , Y and Z are on the circle. The chords XZ and BY intersect at P and $\angle BXY = \angle BPX$.



Prove that

(a) triangle BXY is similar to triangle BPX ,

(b) $BX^2 = BP \times BY$,

(c) XZ is parallel to AC .

[2]

[2]

[2]

1	In $\triangle ACT$ and $\triangle CBT$ $\angle CTA = \angle BTC$ (common angle) $\angle CAT = \angle BCT$ (alternate segment theorem) $\triangle CAT$ and $\triangle CBT$ are similar by AA similarity test $\frac{CT}{BT} = \frac{CA}{BC} = \frac{AT}{CT}$ $CT^2 = AT \times BT$
2	a $\angle CBE = \angle BDC$ (alternate segment theorem) $\angle CDB = \angle ABD$ (alt $\angle$) b Let $\angle BCE = x$ $\angle DCB = 180 - x$ ($\angle$ on st line) $\angle DAB = 180 - (180 - x)$ ($\angle$ in opp seg) $= x$ $\angle BAD = \angle BCE$ $\angle CBE = \angle ABD$ (from part a) $\angle CEB = \angle ADB$ ($\angle$ in triangle) Using AA similarity, triangle CBE is similar to triangle ABD.
3	(i) $\angle ATD = \angle XAB$ (corr angles, $AB \parallel TD$) $= \angle ACB$ (alternate segment theorem) (ii) Since $\angle ATD = \angle ACB$, they fulfil the property of angles in the same segment. $\therefore$ A circle can be drawn passing through the points T, A, D and C. (iii) $\angle ATE = \angle DCE$ [from part (a)] $\angle TEA = \angle CED$ [vert opp $\angle$] $\therefore \triangle ATE$ is similar to $\triangle DCE$ (AA similarity test) $\frac{TE}{CE} = \frac{EA}{ED}$ (Ratio of corr sides of similar triangles are equal) $CE \times AE = DE \times TE$ (proven)
4	(a) $\angle BCA = \angle ACE$ (Common Angle) $\angle ABC = \angle CAE$ (Angles in the alternate segment are equal) $= \angle EAC$ (AC bisects $\angle DAY$) $\triangle BAC = \triangle AEC$ are similar. (b) $\triangle BAC = \triangle AEC$ are similar. $\frac{AC}{EC} = \frac{BC}{AC}$ (Corresponding sides of similar triangle) $AC^2 = EC \times BC$ (Shown)

	(c) $\angle CAY = \angle EAC \text{ (AC bisects } \angle DAY)$ $\angle BAX = \angle EAB \text{ (AB bisects } \angle BAX)$ $\angle BAX + \angle EAB + \angle EAC + \angle CAY = 180^\circ$ $2\angle EAB + 2\angle EAC = 180^\circ$ $\angle EAB + \angle EAC = \angle BAC = 90^\circ$ Since $\angle BAC = 90^\circ$, by angles in a semicircle, BC is the diameter of the circle
5	$\angle QSR = \angle QPR = 60^\circ \text{ (}\angle\text{s in the same segment)}$ $\angle PSR = \angle PQR = 60^\circ \text{ (}\angle\text{s in the same segment)}$ $\angle PSQ = \angle QSR + \angle PSR = 120^\circ$ $\angle QXS = \angle QSX = 60^\circ \text{ (base } \angle\text{s of isosceles triangle)}$ $\angle QXR = 180^\circ - 60^\circ = 120^\circ \text{ (adj. } \angle\text{s on a straight line)}$ $\therefore \angle PSQ = \angle RXQ = 120^\circ$ $\angle QRX = \angle QPS \text{ (}\angle\text{s in the same segment)}$ $PQ = RQ \text{ (equilateral triangle)}$ By AAS congruence test, triangle RXQ is congruent to triangle PSQ. (A1 – AAS) <u>Alternative method using SAS</u> $QS = QX \text{ (given)}$ $PQ = QR \text{ (equilateral triangle)}$ Let $\angle SQP = x$ $\angle SQP = \angle SRP = x \text{ (}\angle\text{s in the same segment)}$ $\angle PQR = \angle PRQ = 60^\circ \text{ (equilateral triangle)}$ $\angle SRQ = \angle PRQ - \angle SRP$ $= 60^\circ - x$ $\angle QSX = \angle QPR = 60^\circ \text{ (}\angle\text{s in the same segment)}$ $\angle QXS = \angle QSX = 60^\circ \text{ (base } \angle\text{s in isosceles triangle)}$ $\angle XQR = \angle QXS - \angle QRX$ $= 60^\circ - (60^\circ - x) \text{ (ext. } \angle\text{s of triangle)}$ $= x$ $= \angle SQP$

6	(a)	To prove $\triangle BXY$ is similar to $\triangle BPX$: $\angle B$ is a common angle $\angle BXY = \angle BPX$ (given) Therefore, $\triangle BXY$ is similar to $\triangle BPX$ (AA)
	(b)	To prove $BX^2 = PB \times BY$: Since $\triangle BXY$ is similar to $\triangle BPX$ from (a), $\frac{BX}{BP} = \frac{BY}{BX}$ $BX^2 = BP \times BY$ (Shown)
	(c)	$\angle CBY = \angle BXY$ (Alt. segment theorem) Since $\angle BXY = \angle BPX$ (given), $\therefore \angle CBY = \angle BPX$ Since $\angle CBY = \angle BPX$, by the converse of alternate angles, XZ is parallel to AC.